

GEOGRAPHY 04 — WEATHERING, MASS MOVEMENT, GROUNDWATER / INDIA EROSION-LANDSLIDES-GROUNDWATER

EXOGENIC SOURCE-TRANSFER-STORAGE SYSTEM — WEATHERING TAXONOMY, CONTROLS, REGOLITH — SOIL EROSION SEQUENCE AND — MASS-MOVEMENT TAXONOMY AND SLOPE-STABILITY — INDIA LANDSLIDE GEOGRAPHY, RISK — GROUNDWATER ARCHITECTURE FROM RECHARGE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE Pending user review source generation g8 and all prior artifacts unchanged

CORE BEFORE EXTRA prior artefacts unchanged exact same master drives poster and tiles

STAGE 00 EXOGENIC SOURCE-TRANSFER-STORAGE SYSTEM

EXOGENIC SOURCE-TRANSFER-STORAGE SYSTEM

EXOGENIC LANDSCAPE SYSTEM

HOW IS ROCK CONVERTED INTO LANDFORM?

ROCK EXPOSED AT SURFACE WEATHERING IN PLACE REGOLITH AND

DENUDATION IS COLLECTIVE LOWERING; WEATHERING, EROSION AND MASS WASTING

EXOGENIC LANDSCAPE SYSTEM

CENTRAL QUESTION: How is rock converted into landform?

ROCK EXPOSED AT SURFACE weathering in place regolith and loose material supplied gravity-driven transfer agent-driven

downslope water, wind, ice, waves erosion + transport deposition temporary storage remobilisation

Denudation is collective lowering; weathering, erosion and mass wasting are linked but distinct processes.

DEFINITION / COMPONENTS

- EXOGENIC LANDSCAPE SYSTEM
- CENTRAL QUESTION: How is rock converted into landform?

TRIGGER / MECHANISM

- ROCK EXPOSED AT SURFACE weathering in place regolith and loose material supplied gravity-driven transfer agent-driven transfer mass wasting
- downslope water, wind, ice, waves erosion + transport deposition temporary storage remobilisation

EFFECT / INTERVENTION

- Denudation is collective lowering; weathering, erosion and mass wasting are linked but distinct processes.
- EXOGENIC LANDSCAPE SYSTEM

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Denudation is a connected source-transfer-storage system, but weathering, erosion and mass wasting must still be distinguished by whether material remains in place, is carried by an agent, or moves mainly under gravity.

STAGE 01 WEATHERING TAXONOMY, CONTROLS, REGOLITH AND SOIL FORMATION

WEATHERING TAXONOMY, CONTROLS, REGOLITH AND SOIL FORMATION

MECHANICAL CHEMICAL BIOLOGICAL THERMAL SOLUTION ROOTS/BURROWS UNLOADING CARBONATION ORGANIC

REGOLITH ORGANIC INPUTS + ORGANISMS + TRANSLOCATION + TIME

R REMOVAL GAIN PARENT ROCK

CONTROLS CONVERGE

MINERALOGY + JOINTS/BEDDING + CLIMATE + RELIEF DRAINAGE +

BLACK SOIL LINK

DECCAN BASALT PARENT, SHRINK-SWELL CLAY, DEEP DRY CRACKS

CLASSIFICATION
MECHANICAL CHEMICAL BIOLOGICAL thermal solution roots/burrows unloading carbonation organic acids frost hydration human exposure salt hydrolysis wet-dry oxidation
REGOLITH organic inputs + organisms + translocation + time

DIAGNOSTIC BASIS

O → A/E → B → C → R removal gain parent rock

PROCESS LINK

Black soil link: Deccan basalt parent, shrink-swell clay, deep dry cracks

EXAM TRAP

MECHANICAL CHEMICAL BIOLOGICAL thermal solution roots/burrows unloading carbonation organic acids frost hydration human exposure salt hydrolysis wet-dry oxidation
REGOLITH organic inputs + organisms + translocation + time

CLASSIFICATION
MECHANICAL CHEMICAL BIOLOGICAL thermal solution roots/burrows unloading carbonation organic acids frost hydration human exposure salt hydrolysis wet-dry oxidation
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DIAGNOSTIC BASIS

- O → A/E → B → C → R removal gain parent rock
- CONTROLS CONVERGE: mineralogy + joints/bedding + climate + relief drainage + organisms + time + land use

PROCESS LINK

- Black soil link: Deccan basalt parent, shrink-swell clay, deep dry cracks
- parent material does not alone determine every soil property.

EXAM TRAP

- MECHANICAL CHEMICAL BIOLOGICAL thermal solution roots/burrows unloading carbonation organic acids frost hydration human exposure salt hydrolysis wet-dry oxidation
- REGOLITH organic inputs + organisms + translocation + time

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Weathering is controlled not by climate alone but by the interaction of energy and moisture with mineralogy, discontinuities, relief, biota, time and land use.

STAGE 02 SOIL EROSION SEQUENCE AND WATERSHED CONTROL LOGIC

SOIL EROSION SEQUENCE AND WATERSHED CONTROL LOGIC

RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION DIFFUSE

CHAMBAL CUE

WATERSHED RESPONSE COVER SOIL

SLOW RUNOFF

SHORTEN SLOPE LENGTH

PROTECT DRAINAGE

TREAT GULLY HEAD

RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION diffuse sheet removal small rill channels entrenched

WATERSHED RESPONSE cover soil → slow runoff → shorten slope length → contour/terrace

protect drainage → treat gully head → rehabilitate catchment

WIND BRANCH surface creep → saltation → suspension

COASTAL BRANCH wave erosion belongs to the coastal-process system; use only when asked.

DEFINITION / COMPONENTS

- RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION diffuse sheet removal small rill channels entrenched gullies ravine landscape: Chambal cue
- WATERSHED RESPONSE cover soil → slow runoff → shorten slope length → contour/terrace

TRIGGER / MECHANISM

- protect drainage → treat gully head → rehabilitate catchment
- WIND BRANCH surface creep → saltation → suspension

EFFECT / INTERVENTION

- COASTAL BRANCH wave erosion belongs to the coastal-process system; use only when asked.
- A check dam alone cannot replace upstream cover, drainage and maintenance.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Rainwater harvesting becomes effective only when capture, water quality, aquifer suitability, maintenance and demand control are treated as one urban water-budget system.

STAGE 03 MASS-MOVEMENT TAXONOMY AND SLOPE-STABILITY MECHANISM

MASS-MOVEMENT TAXONOMY AND SLOPE-STABILITY MECHANISM

MASS MOVEMENT

SLOW SLIDE/FALL FLOW/AVALANCHE CREEP, SOLIFLUCTION ROCKFALL DEBRIS FLOW TRANSLATIONAL

SLOPE BALANCE RESISTING STRENGTH

COHESION + FRICTION + STRUCTURAL SUPPORT VERSUS DRIVING STRESS

GRAVITY COMPONENT + LOADING + TOE LOSS RAIN INFILTRATION

ADDED WEIGHT + PORE PRESSURE

LOWER EFFECTIVE STRENGTH ROAD OR RIVER CUT

MASS MOVEMENT

SLOW SLIDE/FALL FLOW/AVALANCHE creep, solifluction rockfall debris flow translational slide earth or mud flow rotational

snow avalanche

SLOPE BALANCE resisting strength = cohesion + friction + structural support versus driving stress =

CLASSIFICATION

- MASS MOVEMENT
- SLOW SLIDE/FALL FLOW/AVALANCHE creep, solifluction rockfall debris flow translational slide earth or mud flow rotational slump rock/debris avalanche
- snow avalanche

DIAGNOSTIC BASIS

- SLOPE BALANCE resisting strength = cohesion + friction + structural support versus driving stress = gravity component +
- loading + toe loss rain infiltration → added weight + pore pressure → lower effective strength road or
- river cut → steepening / undercutting → support loss quake / quarry / leakage / drainage change →

PROCESS LINK

- trigger or amplification
- FAILURE WHEN DRIVE RESISTANCE
- Material, movement and water define the event: speed alone does not.

EXAM TRAP

- SLOPE BALANCE resisting strength = cohesion + friction + structural support versus driving stress = gravity component +

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Landslides occur at a threshold where driving stress exceeds resisting strength; water is especially important because it can simultaneously add weight, raise pore pressure and weaken material.

STAGE 04 INDIA LANDSLIDE GEOGRAPHY, RISK PRODUCTION AND MITIGATION STACK

INDIA LANDSLIDE GEOGRAPHY, RISK PRODUCTION AND MITIGATION STACK

INDIA LANDSLIDE GEOGRAPHY

YOUNG RELIEF + FRACTURES + INCISION + RAIN +

WESTERN GHATS

ESCARPMENT + DEEP WEATHERING + OROGRAPHIC RAIN + CUTS

WEATHERED SLOPES + ROADS + SETTLEMENTS + DRAINAGE CHANGE

LOSS GEOLOGY/SLOPE RAIN/QUAKE SLIDE/FLOW ROAD/TOWN DISASTER

GSI LADDER INVENTORY

INDIA LANDSLIDE GEOGRAPHY

Himalaya / NE : young relief + fractures + incision + rain + seismicity

Western Ghats : escarpment + deep weathering + orographic rain + cuts

Nilgiris : weathered slopes + roads + settlements + drainage change

SYSTEM / DEFINITION

- INDIA LANDSLIDE GEOGRAPHY
- Himalaya / NE : young relief + fractures + incision + rain + seismicity
- Western Ghats : escarpment + deep weathering + orographic rain + cuts
- Nilgiris : weathered slopes + roads + settlements + drainage change

PROCESS / MECHANISM

- PREDISPOSITION → TRIGGER → MOVEMENT → EXPOSURE → LOSS geology/slope rain/quake slide/flow road/town disaster
- GSI LADDER inventory → susceptibility/hazard study → post-event diagnosis
- monitoring / NLFC support → decision communication
- MITIGATION STACK avoid unsafe siting → manage drainage → protect toe

SPATIAL / INDIA PATTERN

- stabilise slope → bioengineer → control spoil/quarrying
- maintain works → warn/close routes → evacuate and prepare
- inventory maps past events; susceptibility maps relative likelihood
- neither is a deterministic event prediction.

HAZARD / LIMIT / RESPONSE

- Joshimath is a multi-causal subsidence case, not a synonym for landslide.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: India needs a layered landslide strategy—avoid unsafe exposure, control water, stabilise diagnosed slopes, maintain infrastructure, and connect forecasts to last-mile action.

STAGE 05 GROUNDWATER ARCHITECTURE FROM RECHARGE TO WELLS

GROUNDWATER ARCHITECTURE FROM RECHARGE TO WELLS

SOIL MOISTURE

GROUND SURFACE RECHARGE AREA SPRING AT SLOPE

ZONE OF AERATION PERCHED WATER ON LOCAL AQUITARD CAPILLARY

WATER TABLE UNCONFINED AQUIFER PUMPING WELL

rain → infiltration → soil moisture → percolation → recharge

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SYSTEM / DEFINITION

- rain → infiltration → soil moisture → percolation → recharge
- GROUND SURFACE recharge area spring at slope
- ZONE OF AERATION perched water on local aquitard capillary fringe

PROCESS / MECHANISM

- WATER TABLE UNCONFINED AQUIFER pumping well
- rain → infiltration → soil moisture → percolation → recharge
- cone of depression
- AQUITARD slow transmission

SPATIAL / INDIA PATTERN

- CONFINED AQUIFER pressure: water rises to potentiometric level
- AQUICLUDE practically no transmission porosity = fraction of voids
- permeability = connected flow capacity aquifer stores and transmits

HAZARD / LIMIT / RESPONSE

- aquitard transmits slowly artesian water rises under pressure; it flows only if head exceeds ground
- Budget change in storage = recharge + inflow - discharge - pumping - outflow.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Groundwater is a dynamic stock governed by recharge, discharge, pumping, inter-aquifer flow and quality; annual recharge is therefore not a licence to abstract the same quantity everywhere.

STAGE 06 INDIA GROUNDWATER DEPLETION, QUALITY AND SUBSIDENCE GEOGRAPHY

INDIA GROUNDWATER DEPLETION, QUALITY AND SUBSIDENCE GEOGRAPHY

AQUIFER FORM

MAJOR PRESSURE

LAYERED ALLUVIUM

DEPLETION, AS, NITRATE

WEATHERED FRACTURES

DISCONTINUOUS STORAGE

DECCAN BASALT

REGION

Indo-Gangetic
Peninsular
Deccan basalt
Arid hard rock
Coastal belt

AQUIFER FORM

layered alluvium
weathered fractures
contacts + fractures
limited recharge
fresh-saline wedge

MAJOR PRESSURE

depletion, As, nitrate
discontinuous storage
uneven well yield
fluoride, salinity
intrusion and upconing crop-water demand + cheap energy + private wells + urban growth weak aquifer regulation + limited recharge falling head → lower yield → higher lift cost → unequal access compressible sediment: pore-pressure loss + compaction → subsidence

SYSTEM / DEFINITION

- AQUIFER FORM
- MAJOR PRESSURE
- layered alluvium
- depletion, As, nitrate
- weathered fractures

PROCESS / MECHANISM

- Deccan basalt
- contacts + fractures
- uneven well yield
- Arid hard rock
- limited recharge

SPATIAL / INDIA PATTERN

- Coastal belt
- fresh-saline wedge
- intrusion and upconing crop-water demand + cheap energy + private wells + urban growth weak aquifer regulation +
- limited recharge falling head → lower yield → higher lift cost → unequal access compressible sediment: pore-pressure loss
- compaction → subsidence

HAZARD / LIMIT / RESPONSE

- fluoride rock-water interaction nitrate sewage/fertiliser
- salinity coastal or inland
- Quantity and quality require separate evidence, though each limits usable water.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Good geography answers move from process to spatial pattern, then from consequence to a mechanism-matched and institutionally realistic response.

STAGE 07 RECHARGE, DEMAND GOVERNANCE AND INTEGRATED MAINS ANSWER SPINE

RECHARGE, DEMAND GOVERNANCE AND INTEGRATED MAINS ANSWER SPINE

GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS

RECHARGE DEMAND

GOVERNANCE ROOF AND RUNOFF CAPTURE CROP-WATER ALIGNMENT MANAGED AQUIFER

AQUIFER PLAN

CWGB ASSESSMENT + NAQUIM MAPPING + STATE REGULATION

ATAL BHUJAL LEARNING + LOCAL INSTITUTIONS + PUBLIC DATA

ANSWER SPINE

GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS

SUPPLY / RECHARGE DEMAND / GOVERNANCE roof and runoff capture crop-water alignment managed aquifer recharge

treatment well registration/metering spring-shed restoration conjunctive use and reuse quality-safe source water community water budgets

enforceable state/local rules

AQUIFER PLAN

DEFINITION / COMPONENTS

- GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS
- SUPPLY / RECHARGE DEMAND / GOVERNANCE roof and runoff capture crop-water alignment managed aquifer recharge efficient irrigation watershed
- treatment well registration/metering spring-shed restoration conjunctive use and reuse quality-safe source water community water budgets maintenance and monitoring
- enforceable state/local rules
- AQUIFER PLAN

TRIGGER / MECHANISM

- Atal Bhujal learning + local institutions + public data
- ANSWER SPINE
- 1 define stock-flow-quality system
- 2 draw aquifer or India map
- 3 diagnose regional mechanism

EFFECT / INTERVENTION

- 5 match intervention to cause
- 6 quality limits and equity
- Recharge structures are inputs, not proof of water-table recovery.
- Desalination supplies water but does not restore freshwater aquifer head.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Desalination supplies water but does not restore freshwater aquifer head.

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

ADVANCED 1 — EFFECTIVE STRESS, FACTOR OF SAFETY AND

ADVANCED 2 — WEATHERING PROFILES, CLAYS AND HYDRO-MECHANICAL FEEDBACK

ADVANCED 3 — FORECAST SKILL, FALSE ALARMS AND NON-STATIONARITY

ADVANCED 4 — AQUIFER HETEROGENEITY, CAPTURE AND SUSTAINABLE YIELD

ADVANCED 5 — GOVERNANCE EVALUATION

IN TECHNICAL TERMS, SHEAR STRENGTH IS OFTEN REPRESENTED BY

COHESION PLUS EFFECTIVE NORMAL STRESS MULTIPLIED BY FRICTION.

OPTIONAL PROCESS DEPTH

- ADVANCED 1 — EFFECTIVE STRESS, FACTOR OF SAFETY AND THRESHOLDS
- ADVANCED 2 — WEATHERING PROFILES, CLAYS AND HYDRO-MECHANICAL FEEDBACK
- ADVANCED 3 — FORECAST SKILL, FALSE ALARMS AND NON-STATIONARITY
- ADVANCED 4 — AQUIFER HETEROGENEITY, CAPTURE AND SUSTAINABLE YIELD

DATA / INSTITUTION LIMIT

- ADVANCED 5 — GOVERNANCE EVALUATION
- ADVANCED 1 — EFFECTIVE STRESS, FACTOR OF SAFETY AND THRESHOLDS The simple core rule is driving stress versus resisting strength.
- In technical terms, shear strength is often represented by a Mohr-Coulomb relation: cohesion plus effective normal stress multiplied by friction.
- Pore pressure lowers effective normal stress.

CONTEMPORARY PARALLEL.

- A factor of safety above one indicates calculated resistance exceeds driving force; near one means marginal stability.
- Real slopes remain uncertain because geometry, groundwater pathways and material properties vary.
- ADVANCED 2 — WEATHERING PROFILES, CLAYS AND HYDRO-MECHANICAL FEEDBACK Weathering fronts can create saprolite, corestones, clay-rich horizons and
- permeability contrasts.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.